Homework 14 Reflection

Date Display GUI 2

This homework implements a JavaFX-based Date Display GUI application. The main functionality is to display a selected base date, allow the user to choose a date format, and calculate a result date by moving the base date before or after a specified number of days, months, or years. The application starts with the current date using `LocalDate.now()`, supports multiple formats such as `dd/MM/yyyy`, `MM/dd/yyyy`, `dd.MM.yyyy`, and `yyyy-MM-dd`, and provides validation for the amount input.

The application follows the MVC architectural pattern. The `DateModel` class stores the application state, including the selected date, result date, selected date format, and error message. It also contains the date calculation logic using `LocalDate`. The `DateView` class is responsible for creating the JavaFX user interface, including the `DatePicker`, `ComboBox`, `TextField`, radio buttons, labels, and apply button. The `DateController` connects the model and view by setting up listeners, bindings, input validation, and button actions.

Compared with the previous AWT/Swing version, this version is more modular and maintainable because the responsibilities are separated clearly. The previous design could easily mix UI code and application logic, while the new design makes the calculation logic, interface construction, and event handling easier to understand and modify.

JavaFX properties and binding are used to make the interface reactive. The selected date label is bound to both the selected date property and the selected date format property, so it updates automatically when either value changes. The result label is also bound to the result date and selected format. A Boolean binding is used to disable the Apply button when the input amount is invalid, while the error label is bound to the model's error message property.